**Table S1. Search strategy**

| # | Search | Details |
|---|---|---|
| 1 | Vitamin D | "Vitamin D"[Mesh] OR vitamin D*[Text Word] OR "25-hydroxyvitamin D"[Substance] OR "25-hydroxyvitamin D"[Text Word]) |
| 2 | Pregnancy | "Pregnancy"[Mesh] OR "PREGNANT" [Text Word] OR "PREGNANCY" [Text Word] OR "ANTEPARTUM" [Text Word] OR "PRENATAL" [Text Word] OR "ANTE-PARTUM" [Text Word] OR "PRE-NATAL" [Text Word] OR "PREPART" [Text Word] OR "ANTENATAL" [Text Word] |
| 3 | Lactation | "Lactation"[Mesh] OR "Breastfeeding"[Mesh] OR "Breastfeeding" [Text Word] OR "breast feeding" [Text Word] OR "human milk" [Text Word] OR "breast milk" [Text Word] OR "POSTPARTUM" [Text Word] OR "POSTNATAL" [Text Word] OR "POST-PARTUM" [Text Word] OR "POST-NATAL" [Text Word] |
| 4 | Infancy | "Infancy"[Mesh] OR "Infant" [Text Word] OR "PREGNANCY" [Text Word] OR "ANTEPARTUM" [Text Word] OR "PRENATAL" [Text Word] OR "ANTE-PARTUM" [Text Word] OR "PRE-NATAL" [Text Word] OR "PREPART" [Text Word] OR "ANTENATAL" [Text Word] |
| 5 | Neonatal and child growth | "Growth"[Mesh] OR "Growth Disorders"[Mesh] OR "Nutritional Status"[Mesh] OR stunt[Text Word] OR underweight[Text Word] OR wast[Text Word] OR "Body Height"[Mesh] OR "Body Weight"[Mesh] OR "height"[Text Word] OR "weight"[Text Word] OR "Anthropometry"[Mesh] OR Anthropom*[Text Word] |
| 6 | Body composition | " Body composition "[Mesh] OR "muscle, skeletal"[Mesh] OR OR "muscle mass"[Text Word] OR "lean mass"[Text Word] OR "fat mass"[Text Word] OR "adiposity"[Text Word] OR " body mass index"[Text Word] OR "BMI"[Text Word] OR "skinfold thickness"[Text Word] OR "skinfold measurement"[Text Word] |

The search was restricted to human studies in English language. For the search we included #1 AND #2-4 AND #5-6.

**Table S2. Assessment of Bias Risk of Randomized Controlled Trials**

| Study | Sequence generation | Allocation concealment | Blinding of participants and personnel | Blinding of outcome assessment | Incomplete outcome data | Selective outcome data | Other sources of bias |
|---|---|---|---|---|---|---|---|
| Brooke 1980[30] | Unclear | Low | Low | Low | Low | High | High |
| Brustad 2020[40] | Low | Low | Low | Low | Low | Low | Low |
| Cooper 2016[26] | Low | Low | Low | Low | Low | Low | Low |
| Czech-Kowalska 2014[31] | Low | Low | Low | Low | Low | Low | Low |
| Goldring 2013[32] | Low | High | Low | Low | Low | Low | High |
| Hazell 2014[34] | Low | Low | Low | Unclear | High | Low | Low |
| Marya 1988[36] | Unclear | Unclear | Unclear | Unclear | Low | Low | Low |
| Roth 2013[37] | Low | Low | Low | Low | Low | Low | Low |
| Roth 2018[38] | Low | Low | Low | Low | Low | Low | Low |
| Sahoo 2017[27] | Low | Low | Low | Low | High | Low | Low |
| Trilok-Kumar 2015[39] | Low | Low | Low | Unclear | High | Low | Low |
| Vaziri 2016[3] | Low | Low | Low | Low | Low | Low | Low |